

Project Plan: Crypto Simulator

Web-Based Mining Simulation & Virtual Wallet

1. Executive Summary

This document outlines the architecture, design system, and implementation roadmap for a web-based crypto mining simulation game. Inspired by professional mining dashboard applications, this platform will provide a highly immersive, mock-economy experience. Users will navigate a deeply stylized interface to "purchase" virtual mining hardware, manage power output, view simulated rewards, and interact with a mock digital wallet. The entire experience is contained within a web environment, strictly utilizing professional iconography over standard emojis to maintain a premium feel.

2. Technology Stack & Architecture

Layer	Technology	Purpose
Markup	HTML5	Semantic structure for the dashboard, sidebars, and component containers.
Styling	CSS3	Custom styling focusing on deep purples, dark grays, and glassmorphism.
Logic & State	Vanilla JavaScript	Handling mock transactions, miner upgrades, timer countdowns, and state.
Storage	Local Storage (Browser)	Persisting user balance, owned miners, and mock transaction history locally.

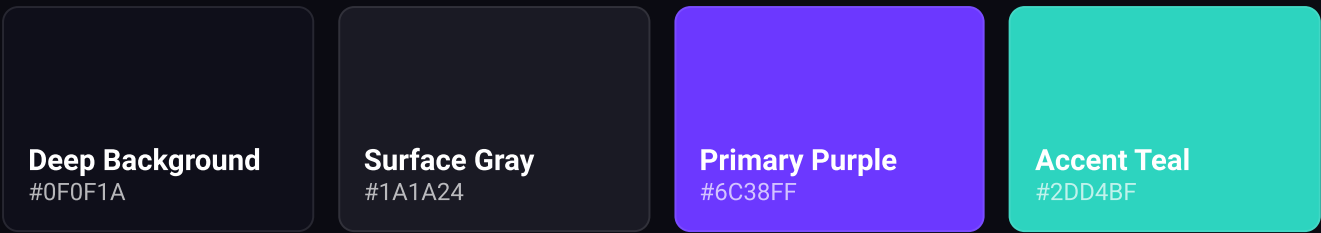
3. Core Concept & Economy

The Simulation Loop

The core gameplay loop involves the user starting with a virtual balance. They navigate to the **Market** to purchase virtual ASICs or mining rigs. Once acquired, these rigs appear in the **My Miners** section and begin generating mock cryptocurrency (e.g., vBTC) over time. The user can then withdraw these mock earnings to their **Wallet**, simulating a real digital asset ecosystem. The interface must reflect real-time metrics such as Terahash per second (TH/s) and Watt efficiency (W/TH), reinforcing the simulation's depth.

4. UI/UX Design System

The visual identity is strictly defined by a dark theme utilizing deep navy/black backgrounds, vibrant purple accents, and metallic gray surfaces. This creates a "cyber" and financial aesthetic suitable for a crypto application.



5. Glassmorphism & Material Guidelines

To achieve the sleek, modern interface requested, the application will heavily utilize glassmorphism for modal windows, floating action bars, and market cards. Specifications include:

- **Background:** Semi-transparent dark overlay (rgba 30, 30, 40, 0.6).
- **Backdrop Filter:** Minimal blur of 10px to 16px to distort underlying elements softly.
- **Borders:** Ultra-thin 1px solid borders using a 10% opacity white (rgba 255, 255, 255, 0.1) to simulate glass edges.
- **Shadows:** Diffused, large spread dark shadows (e.g., 0 8px 32px rgba 0, 0, 0, 0.4) rather than harsh drop shadows.

6. Iconography (Strict No-Emoji Policy)

To maintain a high-end, professional financial simulation, emojis are strictly prohibited across the entire application. All visual indicators will rely on clean, vector-based SVG icons. For example, a wallet will be represented by an outlined leather wallet icon, power output by a lightning bolt vector, and miners by a server-rack silhouette.

7. Screen Breakdown & Navigation

The application is organized into five primary views, accessible via a bottom navigation bar and an expanded side menu for secondary settings.

7.1 Home Dashboard

The central hub for the user. The top section displays the total aggregated wallet balance in large, clear typography. Below this are quick-action circular buttons (Create Miner, Deposit, Earn, Referrals) rendered with glassmorphic backgrounds. The main body features overview cards showing active mining rewards, current TH/s power, and next expected payout countdowns. A marquee or ticker displaying simulated live crypto prices (e.g., BTC, ETH) runs below the main balance.

7.2 Market (Store)

This view operates as the store where users purchase new equipment. The layout features horizontal tabs (Miners, Merch, Listings). The primary view lists visually distinct "Mining Boxes." Each card displays:

- A 3D-style render or high-quality vector of the hardware.
- Key stats in small pill-shaped tags: Power (TH), Efficiency (W/TH), and simulated ROI (%).
- A prominent, full-width "Purchase" button in Primary Purple.

7.3 My Miners (Farm)

This screen is the active management console. It lists all hardware currently owned by the user. Users can click on individual miners to view detailed statistics, initiate artificial "upgrades," or pay simulated maintenance fees. A master dashboard at the top aggregates total power output and average efficiency across the entire virtual farm.

7.4 Wallet & Transactions

Designed to mimic a real Web3 or crypto exchange wallet. The UI shows Total Balance at the top, followed by action buttons: Deposit, Convert, Transfer, and Withdraw. Below, a list itemizes the user's holdings (e.g., Mock-BTC, Mock-USDT). The logic here will allow users to "transfer" funds to other mock usernames, deducting from their local balance and generating a fake transaction hash for immersion.

7.5 Profile & Side Menu

The side menu (accessed via a hamburger icon on the top left) slides out with a glassmorphic blur over the main content. It provides deeper navigation: Account Settings, KYC (mock completion status), Security (2FA simulation), and Support. The profile screen displays the user's VIP Level, ID string, and connected virtual cards.

8. State Management (Vanilla JS Logic)

Because this is a frontend-only simulation, JavaScript will handle the illusion of a live backend:

- **Global State Object:** A central JS object will track `userBalance`, `ownedMiners[]`, and `transactionHistory[]`.
- **Mining Loop:** A `setInterval` function will run every second, calculating rewards based on the aggregate TH/s of the `ownedMiners` array, incrementally updating the virtual pending balance.
- **Storage:** On every significant action (purchase, transfer, or every 60 seconds), the global state is stringified and saved to `localStorage` to ensure the user doesn't lose progress on page refresh.
- **Mock Network Delays:** When a user buys a miner or transfers funds, a `setTimeout` will simulate network latency (1-3 seconds) while showing a CSS spinning loader, enhancing the realistic feel.

9. Implementation Roadmap

The development of this single-page application (SPA) will follow a structured 4-phase approach to ensure layout accuracy and stable simulation logic.

Phase	Focus Area	Key Deliverables
Phase 1	Foundation & Theming	Setup HTML skeleton. Define CSS variables for the deep purple/gray palette. Implement glassmorphism utility classes and integrate SVG icon sets (no emojis).
Phase 2	Layout & Navigation	Build the bottom navigation bar and the slide-out side menu. Create empty container views for Home, Market, Miners, Wallet, and Profile. Implement view-switching logic in JS.
Phase 3	Component Design	Flesh out the internal UI of each view. Build Market cards, Wallet balance displays, and Miner statistic tables to perfectly match the requested visual fidelity.
Phase 4	Simulation Logic	Write the JavaScript state manager. Implement the mock purchasing system, the time-based mining reward intervals, and the fake transaction generation. Tie state to UI updates.